

Introduction to Modeling for Formal Verification

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Today's goals



▼

Module 1

▼

Introduction to Model Checking

Module 2

▼

Modeling

Module 3

▲

Modeling System Composition

Modeling Concurrency Using Transition Systems

✓

Concurrency and Interleaving

Video • 10 min

✓

Concurrency and Interleaving: Examples

Video • 8 min

✓

Principles of Model Checking - Section 2.2.1, (Page 35-39)

Reading • 30 min

✓

Modeling Concurrency Using Transition Systems Quiz

Graded Assignment • Grade: 100%

Modeling Communications Via Shared Variables

✓

Interleaving Program Graphs

Video • 11 min

✓

Critical Actions and Atomicity

Video • 10 min

✓

Peterson's Mutual Exclusion Algorithm

Video • 12 min

✓

Principles of Model Checking - Section 2.2.2, (Page 39-47)

Reading • 30 min

●

Modeling Communications Via Shared Variables Quiz

Graded Assignment • 20 min

Modeling Communications Via Handshaking

●

Synchronization and Handshaking

Video • 12 min

●

Example: Railroad Crossing

Video • 13 min

●

Principles of Model Checking - Section 2.2.3, (Page 48-52)

Reading • 30 min

●

Modeling Communications Via Handshaking Quiz

Graded Assignment • 10 min

Synchronous Parallelism

●

Synchronous Product - Part 1

Video • 10 min

●

Synchronous Product - Part 2

Video • 10 min

●

The State Space Explosion Problem

Video • 9 min

●

Principles of Model Checking - Section 2.2.6, (Page 73-75)

Reading • 20 min

●

Synchronous Parallelism Quiz

Graded Assignment • 20 min

# Principles of Model Checking - Section 2.2.2, (Page 39-47)

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## Textbook - Principles of Model Checking - Chapter 2.2.2 (page 39-47)

Section 2.2.2 introduces the communication model via shared variables. This communication model can be found in multithreaded applications where threads in a program communication with each other by accessing shared memory, which are modeled as shared variables. This section shows that interleaving the transition systems of the consisting processes to form the transition system model for the system composed of those processes does not lead to valid result. Instead, the program graph for the system is found first by interleaving the program graphs of the consisting processes, and the transition system model for the whole system is derived from the constructed program graph. Additionally, the concepts of critical actions and atomicity are explained.

Go to next item

✓ Completed

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